

QUANTUM BRANCHED FLOW: COHERENCE GRAPH
DYNAMICS
AND THE SPECTRAL GEOMETRY OF
DECOHERENCE

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ABSTRACT. We develop a two-layer graph framework for quantum decoherence in which branch formation is identified with coherence graph fragmentation. Starting from the von Neumann equation alone, we derive two objects with distinct physical roles. The coupling graph G_H encodes the partition structure the Hamiltonian imposes on diagonal amplitude dynamics: an edge exists between basis states $|i\rangle$ and $|k\rangle$ if and only if $H_{ik} \neq 0$. The coherence graph $G_\rho(t)$ encodes the current off-diagonal density matrix elements and evolves dynamically under environmental decoherence. A flow current $J_{i \rightarrow k} = (2/\hbar)\text{Im}(H_{ik}\rho_{ki})$, derived directly from the von Neumann equation, governs the redistribution of diagonal amplitude weight. As decoherence suppresses inter-sector coherence weights, the flow current between sectors vanishes and amplitude sectors become dynamically isolated subgraphs — branch sectors.

The framework draws a structural correspondence with classical branched flow, in which persistent amplitude channels form spontaneously when waves propagate through weakly disordered media. In the quantum setting, G_H plays the role of the background medium and $G_\rho(t)$ plays the role of the wave field. Branch sectors are the persistent channels, and their locations are latent in the spectral geometry of G_H : the low-eigenvalue eigenvectors of the graph Laplacian $L(G_H)$ — in particular the Fiedler vector — predict branch sector assignments exactly, confirmed numerically across 250 block-structured Hamiltonians with perfect alignment. This prediction is conditional on two premises: the Hamiltonian must have block-structured coupling topology ($H_{\text{inter}}/H_{\text{intra}} \lesssim 0.65$), and the environment must couple selectively to inter-sector coherences ($\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$). Both conditions are satisfied in any strong-measurement regime and are physically motivated by einselection; neither is derived from the Hamiltonian alone. Branch formation is a spectral transition: new near-zero eigenvalues appear in $L(G_\rho(t))$ as sectors form, with 91.3% raw agreement between spectral and topological fragmentation measures (95.8% with spectral threshold calibrated via the complete bipartite graph $K_{m,m}$; see Section 9 and [1]).

Explicit results include: fringe visibility in the double-slit experiment equals the inter-path coherence weight $|\rho_{LR}(t)|$ exactly at every stage of decoherence; the maximum Bell violation for a partially dephased singlet is $S_{\text{max}} = 2\sqrt{1 + V^2}$ where V is the normalized coherence weight; and eigenvalue shifts under approximate decoherence scale as $O(\varepsilon^{1.113})$ with dynamic restoration to stable sector structure confirmed globally. The spectral gap λ_1 of $L(G_H)$ governs the regime of sector structure that forms rather than formation timescales, which are dominated by the decoherence rate γ . Key open problems — basis selection, temporal stability, and the Born rule — are identified and precisely located.

1. INTRODUCTION

When waves propagate through a weakly disordered medium — light through a slightly irregular glass, electrons through a disordered conductor, sound through turbulent water — they do not spread diffusely. Instead they self-organize into narrow, persistent channels: branched flow [8, 9]. The channels form spontaneously, persist over long distances, and their statistical properties depend on coarse features of the medium rather than fine-grained details. The phenomenon is universal across wave systems and is governed by the spectral geometry of the background potential through which the waves propagate.

This paper develops an analogous framework for quantum decoherence. When a quantum system interacts with its environment, its coherence structure does not decay uniformly. It fragments — organizing into persistent sectors whose locations are latent in the spectral geometry of the Hamiltonian. We call this quantum branched flow, and we develop a precise geometric framework for it based on two graph objects derived directly from the von Neumann equation.

The coupling graph G_H encodes the partition structure the Hamiltonian imposes on diagonal amplitude dynamics: nodes are basis states, edges connect pairs (i,k) with $H_{ik} \neq 0$. This graph plays the role of the background medium in classical branched flow — it is fixed, does not evolve with the quantum state, and its spectral geometry determines where persistent channels will form. The coherence graph $G_\rho(t)$ encodes the current off-diagonal density matrix elements and evolves dynamically. It plays the role of the wave field. Branch sectors — the dynamically isolated subgraphs that emerge as decoherence suppresses inter-sector coherence weights — are the persistent channels of quantum branched flow.

The structural analogy is productive, though it should be understood as a guiding correspondence rather than a formal identification. Classical branched flow involves stochastic disorder and wave focusing; the quantum setting has neither. What transfers is the mathematical machinery — spectral graph theory [12], Cheeger inequalities [11], caustic analysis — and the organising insight that persistent channels are latent in a fixed background structure. The low-eigenvalue eigenvectors of the graph Laplacian $L(G_H)$ predict branch sector assignments exactly, confirmed numerically across 250 block-structured Hamiltonians with perfect alignment between Fiedler vector predictions and simulated sector structure. Branch formation is a spectral transition: new near-zero eigenvalues appear in $L(G_\rho(t))$ as sectors form (see Section

9 and [1] for quantitative agreement between spectral and topological fragmentation measures).

The framework makes no modifications to quantum formalism and introduces no new physical postulates. Every result follows from the von Neumann equation and standard decoherence theory [5, 6, 7]. It has natural application to the interpretation of quantum mechanics — in particular to the Everettian program [3, 4], where the branch structure emerging from coherence graph fragmentation provides a geometric grounding for the notion of branch identity across time — but the framework and its results do not depend on any interpretational commitment. The spectral geometry of decoherence is physically real regardless of one’s preferred interpretation. This paper is the first in a series developing the CGA framework; companion papers in preparation address the open problems identified in Section 8.

What this paper establishes

The flow current $J_{i \rightarrow k} = (2/\hbar)\text{Im}(H_{ik}\rho_{ki})$ follows directly from the von Neumann equation as the rate of change of diagonal amplitude weights. It is not postulated.

The coupling graph G_H is the partition structure the von Neumann equation imposes on diagonal dynamics — edges exist between nodes i and k if and only if $H_{ik} \neq 0$, because those are precisely the pairs that contribute to amplitude flow. This is derived in Section 2.

Branch formation as coherence graph fragmentation is the geometric consequence of environmental decoherence suppressing inter-sector coherence weights in $G_\rho(t)$. The flow current between sectors vanishes when those weights vanish — exactly, not approximately.

The framework reproduces known quantum results exactly when expressed in coherence graph language, providing consistency checks rather than new predictions. Fringe visibility in the double-slit experiment equals the inter-path coherence weight $|\rho_{LR}(t)|$ directly. The maximum Bell violation for a partially dephased singlet is $S_{\max} = 2\sqrt{1 + V^2}$ where V is the normalized inter-photon coherence weight — derived from the Horodecki criterion and confirmed numerically to machine precision. These relationships are not new physics; they are standard results that the coherence graph formulation recovers exactly (Section 6). The framework’s contribution is not new experimental predictions but a structural decomposition — the two-layer graph structure, the flow current, and the Fiedler partition criterion — that makes the geometric content of decoherence explicit and computable.

Open problems this framework raises

Basis selection is required as an input. The derivation of G_H from the von Neumann equation holds in any basis, but different bases yield different graphs. Zurek’s einselection program [5] provides the physically motivated candidate — the pointer state basis — and Paper 3 of this series establishes the formal equivalence between the CGA basis criterion and the predictability sieve unconditionally at $t = 0$ and in the secular regime at finite times. The residual open question is the exact equivalence at finite times outside the secular regime, located at Section 8.1.

Temporal stability of branch sectors is established in Paper 2 via spectral methods (KMS symmetry, Cheeger inequality, and Davis-Kahan). The spectral structure is developed in Section 5, identifying the Liou-villian gap and Cheeger constant of G_H as the key quantities. Section 8.2 notes the relationship between the Gronwall sufficient condition derived here and the complete spectral result of Paper 2.

The Born rule is not derived here. The diagonal weights ρ_{ii} are treated as mathematical quantities without assuming their interpretation as probabilities. The question of what grounds that interpretation is an open problem precisely located at Section 8.4 and addressed in Papers 4 and 5 of this series.

Scope

The framework is developed within non-relativistic quantum mechanics. Spin and other internal degrees of freedom are included naturally: basis states $|i\rangle$ carry whatever quantum numbers are appropriate to the system, and the derivation is unchanged. Extension to relativistic quantum field theory is left for future work.

Motivation

The standard account of decoherence is formulated in terms of the density matrix $\rho(t)$ and its off-diagonal elements. Environmental interaction drives $|\rho_{ij}(t)| \rightarrow 0$ for certain pairs (i, j) , and the system is said to have decohered. This account is correct as far as it goes. What it does not provide is a structural answer to the question of *where* coherence suppression occurs — which pairs (i, j) decohere, in what order, and with what relationship to the underlying Hamiltonian.

The density matrix evolves as a whole. There is no natural decomposition of $\rho(t)$ into a static background structure and a dynamical field, and therefore no natural language for asking which features of the Hamiltonian determine which coherences survive. The sector structure that emerges under decoherence is visible in $\rho(t)$ after the fact, but the density matrix formalism does not make it easy to ask — let alone answer — where that structure was latent before decoherence acted.

This paper introduces such a decomposition. The von Neumann equation itself, when its diagonal and off-diagonal dynamics are examined separately, generates two objects with distinct physical roles: a static coupling graph G_H encoding the Hamiltonian's partition structure, and a dynamical coherence graph $G_\rho(t)$ encoding the current coherence state. The decomposition is not imposed; it is derived. The two-layer structure makes it possible to ask, and answer precisely, which features of the Hamiltonian determine the sector partition — and the answer is spectral: the Fiedler vector of $L(G_H)$ determines where coherence graph fragmentation will occur, independent of the fine-grained dynamics that drive it.

The practical consequence is a computable criterion for branch structure that applies before decoherence has run to completion: given the Hamiltonian, the sector partition is determined. This is the sense in which branch structure is latent in G_H . The decoherence process does not create the partition; it reveals it (see Section 4.2 for the precise distinction between partition and formation).

2. FROM THE VON NEUMANN EQUATION TO THE COHERENCE GRAPH

The coherence graph is not imposed on quantum mechanics as an interpretive layer. It is the structure the von Neumann equation itself generates when its diagonal and off-diagonal dynamics are examined separately. This section derives that structure from first principles, establishing why the Hamiltonian matrix elements H_{ij} play the role they do, and precisely locating the one input the derivation requires that is not itself derived.

2.1. Starting point: the von Neumann equation. The density operator $\rho(t)$ of a closed quantum system evolves under the von Neumann equation [2]:

$$(1) \quad i\hbar d\rho/dt = [H, \rho]$$

This is exact and fully equivalent to the Schrödinger equation for pure states. No additional assumptions are made. We work within non-relativistic quantum mechanics; the Hilbert space $\mathcal{H}$ is finite or countably infinite dimensional.

Choose an orthonormal basis $\{|i\rangle\}$ for $\mathcal{H}$. The density matrix elements are $\rho_{ij}(t) = \langle i|\rho(t)|j\rangle$. The von Neumann equation in this basis gives:

$$(2) \quad i\hbar d\rho_{ij}/dt = [H, \rho]_{ij} = \sum_k (H_{ik}\rho_{kj} - \rho_{ik}H_{kj})$$

2.2. Diagonal elements and the flow current. For diagonal elements $i = j$, the derivation gives:

$$(3) \quad d\rho_{ii}/dt = (2/\hbar) \sum_k \text{Im}(H_{ik}\rho_{ki}) = \sum_k J_{k \rightarrow i}$$

where the flow current from node k to node i is:

$$(4) \quad J_{k \rightarrow i} = (2/\hbar) \text{Im}(H_{ik}\rho_{ki})$$

This is not a definition imported from outside — it is what the von Neumann equation gives for the rate of change of diagonal elements. Two properties follow immediately: $J_{k \rightarrow i} = -J_{i \rightarrow k}$ (antisymmetry) and $\sum_i d\rho_{ii}/dt = 0$ (conservation of total diagonal weight). A direct consequence: tunneling is sparse amplitude flow along weakly coupled edges — $J_{i \rightarrow k}$ is small when $|H_{ik}|$ is small but nonzero whenever $H_{ik} \neq 0$ and $\rho_{ki} \neq 0$.

2.3. The partition structure: why H element-wise. Examining the flow current between a specific pair (i, k) : when $H_{ik} = 0$, $J_{k \rightarrow i} = 0$ exactly — not approximately, not negligibly, but identically zero regardless of ρ_{ki} . When the Hamiltonian matrix element between two nodes is zero, those nodes cannot exchange amplitude under any circumstances.

This observation determines the coupling graph construction. The von Neumann equation partitions diagonal dynamics according to the zero and nonzero structure of H in the chosen basis: an edge (i, k) belongs to E_H if and only if $J_{k \rightarrow i}$ can be nonzero, which holds if and only if $H_{ik} \neq 0$. The topological structure of G_H is therefore fixed by the Hamiltonian matrix structure in the chosen basis — not a modelling choice made on top of the physics, but a direct consequence of which pairs of states can exchange amplitude at all.

The edge weights of G_H admit two natural choices depending on context, and it is important to distinguish them. The first is $w_{ik} = |H_{ik}|$, which measures the magnitude of the Hamiltonian matrix element directly. This is the natural choice for Fiedler vector analysis: the Fiedler vector of the graph Laplacian $L(G_H)$ with these weights identifies the partition that minimises inter-sector coupling, which is exactly the sector assignment problem. The second is $w_{ik} = 2|H_{ik}|^2/(\hbar\gamma_{ik})$, which is

the effective diffusion rate for diagonal amplitude under the secular approximation (derived in Section 5.2). This is the natural choice for studying the time evolution of populations in the secular regime. Both weightings share the same edge set — the same topological structure derived above — and therefore the same zero/nonzero partition structure. The distinction between them does not affect which nodes are coupled; it affects the quantitative strength assigned to each coupling. Throughout this paper, Definition 3.1 uses $w_{ik} = |H_{ik}|$ for the Fiedler analysis; Section 5.2 uses the secular-regime weights for the Laplacian flow equation.

2.4. Off-diagonal elements and the coherence weights. For off-diagonal elements $i \neq j$, environmental decoherence acts specifically on these elements — suppressing $|\rho_{ij}(t)|$ for pairs in different branch sectors while leaving the diagonal elements ρ_{ii} and the coupling structure G_H intact. The coherence graph $G_\rho(t)$ encodes the current values of $|\rho_{ij}(t)|$ as edge weights. Unlike G_H it is dynamical. The flow current depends on both layers simultaneously: when either H_{ik} or ρ_{ki} vanishes, the flow current vanishes.

2.5. The one required input: basis selection. The derivation establishes the coherence graph construction given a basis $|i\rangle$. The one input not provided by the derivation is the choice of basis. Different bases yield different coupling graphs and different apparent flow structures. Zurek’s einselection program [5] identifies the pointer state basis as the physically motivated candidate — the basis selected by the structure of the system-environment interaction, stable under environmental entanglement. Formally grounding the framework in that basis is developed further in Section 3.5.

3. THE TWO-LAYER GRAPH STRUCTURE

The derivation of Section 2 established two distinct objects with distinct physical roles. Keeping them separate is essential for reasoning clearly about what decoherence does and does not do.

3.1. The coupling graph G_H .

Definition 3.1 (Coupling graph). Let $\{|i\rangle\}$ be a fixed orthonormal basis for the Hilbert space $\mathcal{H}$ and let H be the Hamiltonian. The coupling graph $G_H = (V, E_H, w_H)$ has node set $V = |i\rangle$, edge set $E_H = (i,j) : H_{ij} \neq 0$, and edge weights $w_{ij} = |H_{ij}|$. The edge set is derived: as shown in Section 2.3, edge $(i,j) \in E_H$ if and only if $J_{j \rightarrow i}$ can be nonzero, which requires $H_{ij} \neq 0$. The edge weights $w_{ij} = |H_{ij}|$ are

the natural choice for Fiedler analysis of partition structure. Section 5.3 derives an alternative weighting $w_{ij} = 2|H_{ij}|^2/(\hbar\gamma_{ij})$ appropriate for the secular-regime flow equation; both weightings share the same edge set.

G_H is a fixed property of the physical system. It does not evolve with the quantum state. Its topology — which nodes are coupled and with what strength — is the static background structure through which quantum amplitude navigates.

3.2. The coherence graph $G_\rho(t)$.

Definition 3.2 (Coherence graph). The coherence graph $G_\rho(t) = (V, E_\rho(t), w_\rho(t))$ has the same node set $V = \{|i\rangle\}$ as G_H . Its edge weights are $w_{ij}(t) = |\rho_{ij}(t)|$, and an edge (i, j) is present when $|\rho_{ij}(t)| > 0$. The diagonal terms $\rho_{ii}(t)$ are carried as node weights.

Unlike G_H , the coherence graph is dynamical. Environmental decoherence acts specifically on $G_\rho(t)$ — suppressing off-diagonal weights $|\rho_{ij}(t)|$ between certain node pairs while leaving the diagonal weights $\rho_{ii}(t)$ and the structure of G_H intact.

3.3. Amplitude flow as a two-layer function. The flow current $J_{i \rightarrow k} = (2/\hbar)\text{Im}(H_{ik}\rho_{ki})$ depends on both graphs simultaneously. Three physically distinct cases follow:

$H_{ik} \neq 0, \rho_{ki} \neq 0$: Coupled and coherent. Flow occurs.

$H_{ik} \neq 0, \rho_{ki} \approx 0$: Coupled but decohered. No flow despite physical coupling. This is the branch-formed state.

$H_{ik} = 0$: Uncoupled. No flow possible.

The second case is the physically important one. Decohered branch sectors are not physically isolated from one another — the Hamiltonian retains nonzero matrix elements between states in different sectors — but they are dynamically isolated, because the coherence ρ_{ki} required for nonzero flow current has been suppressed. This distinction cannot be stated in a single-layer account.

3.4. Branch formation as a Layer 2 process. Branch formation is the fragmentation of $G_\rho(t)$ into disconnected components. This is precisely a Layer 2 process. As environmental decoherence proceeds, off-diagonal weights $|\rho_{ij}(t)|$ between states in different sectors are suppressed toward zero. In the idealized limit of perfect decoherence these weights vanish exactly and $G_\rho(t)$ fragments into disconnected components — the branch sectors.

The physically interesting case is when G_H is connected — when every pair of nodes is linked by some path of nonzero H_{ij} . Branch

sectors are dynamically emergent features of $G_\rho(t)$: connected components that appear as decoherence suppresses inter-sector coherence weights toward zero. The low-eigenvalue eigenvectors of $L(G_H)$ identify where those components will form — the latent sector structure — but the sectors themselves exist in $G_\rho(t)$, not G_H .

3.5. Connection to pointer states and einselection. The two-layer structure connects directly to Zurek’s einselection program [5, 6]. In the geometric language developed here, pointer states are nodes in G_H whose coherence graph edges to environmental nodes are suppressed preferentially by the system-environment interaction. This connection does more than suggest a basis choice; it provides the physical justification for the environment condition (C3: $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$) that the branch formation results require.

The argument runs as follows. The Lindblad equation for an open system S coupled to environment E takes the form $\dot{\rho} = -i[H_S + H_{SE}, \rho]/\hbar + \mathcal{D}[\rho]$, where the dissipator $\mathcal{D}$ encodes the environment’s action. For a local environment — one that acts on spatially or energetically separated subsystems independently — $\mathcal{D}$ couples preferentially to operators that are diagonal in the pointer basis: the basis selected by H_{SE} as stable under monitoring [5]. In that basis, the dephasing rates γ_{ij} between pointer states within a sector (intra-sector coherences) are suppressed relative to rates between states in different sectors (inter-sector coherences), precisely because pointer states are the eigenstates of the system operator that H_{SE} measures. The condition $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$ is therefore not an independent assumption but a consequence of defining the basis via einselection: it holds whenever the pointer basis is the one in which the coherence graph is constructed.

C3 is therefore a physical premise, not a free parameter: it is motivated by the structure of local system-environment interactions and is consistent with the standard decoherence literature [6, 7]. It is not derived from first principles in this paper. The argument above shows that C3 is the *expected* consequence of working in the pointer basis, but establishing this rigorously requires showing that (a) the einselected basis exists for the system class in question, and (b) the resulting dephasing rates satisfy the quantitative threshold needed for the formation results. The first is the content of Zurek’s einselection program [5]; the second is addressed in Paper 3 of this series. What is established here is the conditional structure: given a basis in which C3 holds, the framework’s predictions follow. The basis selection problem and the environment condition problem are therefore the same problem,

and grounding the construction in the einselected basis resolves both simultaneously — but that grounding is assumed here, not derived.

3.6. System-apparatus Hamiltonians and the block-structured graph. The block-structured Hamiltonians used throughout this paper and in the numerical suite are not purpose-built test cases. They are the general form produced by any system-apparatus interaction of the von Neumann measurement type. This connection is worth making explicit, because it determines which physical systems the framework applies to and why.

Consider a composite system $S \otimes A$, where S has d_S states $|s_k\rangle$ and the apparatus A has d_A states $|a_j\rangle$. The total Hamiltonian is

$$(5) \quad H = H_S \otimes I_A + I_S \otimes H_A + H_{\text{int}}.$$

For a von Neumann measurement interaction,

$$(6) \quad H_{\text{int}} = \sum_k |s_k\rangle\langle s_k| \otimes F_k,$$

where F_k is the apparatus operator activated when the system occupies state $|s_k\rangle$ [6]. The matrix elements of H in the product basis $|s_k, a_j\rangle$ split cleanly by type. Intra-sector elements (same system state k , different apparatus states $j \neq m$) receive contributions from both H_A and H_{int} :

$$(7) \quad \langle s_k, a_j | H | s_k, a_m \rangle = \langle a_j | H_A + F_k | a_m \rangle.$$

Inter-sector elements (different system states $k \neq l$, same apparatus state $j = m$) receive contributions only from H_S :

$$(8) \quad \langle s_k, a_j | H | s_l, a_j \rangle = \langle s_k | H_S | s_l \rangle.$$

Cross-type elements ($k \neq l$, $j \neq m$) vanish: H_{int} is block-diagonal in the system index and contributes no edges between nodes in different system-state sectors.

The coupling graph G_H therefore has exactly the block-structured topology: d_S sectors of size d_A each, with dense intra-sector edges from $H_A + F_k$ and sparse inter-sector edges of weight $|\langle s_k | H_S | s_l \rangle|$ connecting only same-apparatus-state node pairs across sectors. For $d_S = 2$ (qubit system) and $d_A \gg 1$ (large apparatus), this is the two-clique block structure. The branching condition $H_{\text{inter}}/H_{\text{intra}} \lesssim 0.3$ is a sufficient condition for robust formation at $\gamma_{\text{inter}}/\gamma_{\text{intra}} = 50$; the empirical transition from high to low formation rate occurs near $H_{\text{inter}}/H_{\text{intra}} \approx 0.65$ (Result 24). This maps to $|\langle s_k | H_S | s_l \rangle|/\|H_A + F_k\| \lesssim 0.65$ — the condition that the system Hamiltonian is weak relative to the apparatus dynamics, satisfied in any strong-measurement regime.

The 250 random Hamiltonians in the numerical suite therefore test robustness to variation in the matrix elements of H_A and F_k , which are unconstrained by the measurement structure. The topology is not a choice; it is a consequence of the Hamiltonian decomposition.

Definition 3.3 (Block-structured coupling graph). G_H is *block-structured* with blocks A and B if: (i) A and B partition the node set, with $|A| = |B| = d/2$; (ii) the induced subgraphs $G_H|_A$ and $G_H|_B$ are connected; and (iii) the inter-block edges (connecting A to B) carry coupling weight $H_{\text{inter}} \ll H_{\text{intra}}$ relative to the intra-block edges. The inter-block edge set $\partial(A, B) = \{(i, j) : i \in A, j \in B, H_{ij} \neq 0\}$ is the block boundary.

Systems that do not have this structure produce different G_H topologies, and the Fiedler analysis correctly predicts they will not exhibit two-sector branching. Single-site spin systems have no large apparatus subsystem. Coherently-coupled qubit-cavity models (Jaynes-Cummings) use $H_{\text{int}} = g(a\sigma_+ + a^\dagger\sigma_-)$, which is not of the von Neumann form (6) and generates inter-sector edges that preclude clean sector formation. Nearest-neighbour spin chains have path-graph G_H topology with no dense intra-sector subgraph. The Phase 2B negative results on path, cycle, and star graph topologies confirm that these systems do not produce stable two-sector structure. These are correct predictions about which physical systems branch and which do not, not limitations of the framework.

4. BRANCH FORMATION AS COHERENCE GRAPH FRAGMENTATION

4.1. The branch formation criterion. Environmental decoherence suppresses off-diagonal coherence weights between dynamically evolving components. As decoherence proceeds:

$$(9) \quad \rho_{ij}(t) \rightarrow 0 \text{ for } i \in A, j \in B \quad \Rightarrow \quad J_{A \leftrightarrow B} \rightarrow 0$$

Definition 4.1 (Branch sector). A branch sector is a connected component of $G_\rho(t)$ that has become dynamically isolated from other components through the suppression of inter-component coherence weights. Formally, regions A and B constitute distinct branch sectors at threshold $\varepsilon > 0$ when $|\rho_{ij}(t)| < \varepsilon$ for all $i \in A, j \in B$, and the resulting inter-sector flow current satisfies $|J_{A \leftrightarrow B}| \leq 2\|H_{\text{inter}}\|_F \varepsilon / \hbar$ (from equation (4)). The threshold ε is small relative to the intra-sector coherence scale: $\varepsilon \ll \max_{i,j \in A} |\rho_{ij}(t)|$.

4.2. What decoherence does and does not do. The *partition* — which nodes belong to which sector — is determined by G_H alone (Theorem 5.2), and in this precise sense the sector structure is latent in the Hamiltonian before decoherence acts. But *formation* — whether sectors actually emerge as dynamically isolated components — requires the environment (Theorem 5.1). Before decoherence acts, $G_\rho(t)$ is fully connected and amplitude flows freely between what will become distinct sectors. Decoherence acts on $G_\rho(t)$ by suppressing the off-diagonal coherence weights $|\rho_{ij}(t)|$ between nodes in different sectors, while the Hamiltonian coupling between sectors persists unchanged. The shorthand “decoherence reveals latent structure” is therefore precise for the partition but not for formation: the environment is an active participant in whether the latent partition is realised as dynamical isolation.

4.3. Idealized and approximate decoherence. The branch formation criterion holds exactly in the idealized limit of perfect decoherence. In realistic physics residual off-diagonal terms persist and the flow current between sectors is suppressed but not identically zero. A sufficient condition for stability is: $\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$, with residual inter-sector coherence bounded by $2\|H_{\text{inter}}\|_F/(\hbar\gamma_{\text{inter}} - 2\|H_{\text{intra}}\|_{\text{op}})$. The spectral stability theorem of Paper 2 (Paper 2, Section 5, Theorem 5.2) gives the complete result in the secular regime.

Environment condition (premise, not consequence). The framework requires that the environment couples selectively to inter-sector coherences: $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$. Under uniform dephasing (equal γ on all coherences) the intra-block and inter-block coherences decay at essentially the same rate, no stable two-sector intermediate forms, and the system passes directly to singleton fragmentation. This is confirmed numerically. The environment condition is the physically realistic case for local environments acting on spatially or energetically separated subsystems; it is stated here as an explicit premise of the framework.

4.4. Branch identity across time. A persistent branch sector is a stable flow channel — a region of $G_\rho(t)$ whose internal connectivity is maintained and whose external coherence links remain suppressed over time. The stability condition is a competition between the inter-sector dephasing rate γ_{inter} and the intra-sector Hamiltonian spectral norm $\|H_{\text{intra}}\|_{\text{op}}$. When $\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$, inter-sector coherences are bounded for all time. The relationship between this sufficient condition and the empirical stability boundary is characterised in Section 8.2.

4.5. The quantum-to-classical transition. Before significant decoherence, the system exhibits quantum behavior — amplitude propagates through the channel structure of G_H , but $G_\rho(t)$ remains sufficiently connected that channels retain nonzero coherence weights between them. As decoherence drives inter-sector coherence weights toward zero, the system crosses into classical behavior. The transition is continuous rather than discrete.

5. THE BRANCHED FLOW CORRESPONDENCE

5.1. Classical branched flow. Classical branched flow is a phenomenon in wave and ray physics in which transport through a weakly disordered medium forms narrow, persistent channels rather than spreading diffusively [8, 9, 10]. Its defining features are stable flow channels whose statistics depend on coarse properties of the medium, and the role of caustics — focusing singularities in the ray flow — in determining where channels form.

5.2. The structural correspondence. The two-layer graph structure maps onto classical branched flow with structural precision:

G_H plays the role of the background medium — fixed, does not evolve with the quantum state.

$G_\rho(t)$ plays the role of the wave field. Branch sectors are the persistent channels of quantum branched flow.

- Low-eigenvalue structure of $L(G_H)$ plays the structural role of caustics — the spectral geometry of the background determines where persistent channels concentrate, in both the classical and quantum settings.
- Channel isolation by disorder corresponds to isolation by decoherence suppressing edge weights.

The caustic analogy is structural rather than quantitative: in classical branched flow, caustics are ray-focusing singularities where intensity diverges; in the quantum setting, the low-eigenvalue eigenvectors of $L(G_H)$ identify where the partition forms, but there is no direct quantum analog of intensity divergence. What transfers is the organising principle — persistent channel locations are encoded in the spectral geometry of the background — not the specific focusing mechanism. The correspondence does not generate new predictions beyond what the spectral graph theory provides directly; its value is organisational, connecting a well-studied classical phenomenon to the quantum decoherence setting and motivating the spectral tools (Fiedler analysis, Cheeger constants) that do the analytic work.

In the secular approximation — valid when dephasing rates γ_{ij} are large relative to Hamiltonian transition frequencies (quantitatively: $\gamma/H_{\max} \gtrsim 5$ for $\pm 10\%$ relative error, with the approximation breaking down substantially for $\gamma/H_{\max} \lesssim 2$, confirmed numerically) — coherences are slaved to the populations:

$$(10) \quad \rho_{ij} \approx -iH_{ij}(\rho_{ii} - \rho_{jj})/(\hbar\gamma_{ij}), \quad i \neq j$$

Substituting into the flow current gives the graph Laplacian equation of motion:

$$(11) \quad d\rho_{ii}/dt = \sum_k w_{ik}(\rho_{kk} - \rho_{ii}) = -(L\rho^{\text{diag}})_i$$

where $w_{ik} = 2|H_{ik}|^2/(\hbar\gamma_{ik})$, ρ^{diag} denotes the vector of diagonal elements $(\rho_{11}, \dots, \rho_{nn})$, and L is the graph Laplacian of G_H . These secular-regime weights differ from the edge weights $|H_{ik}|$ of Definition 3.1, but share the same edge set: $w_{ik} \neq 0$ if and only if $H_{ik} \neq 0$. The two weightings answer different questions — $|H_{ik}|$ is appropriate for Fiedler partition analysis (which nodes couple to which); $2|H_{ik}|^2/(\hbar\gamma_{ik})$ is appropriate for population dynamics in the secular regime (how fast amplitude flows). Both are derived from G_H 's edge structure, not imposed on it. The identification of diagonal weight evolution with $-(L\rho^{\text{diag}})_i$ is reliable when $\gamma/H_{\max} \gtrsim 5$, where $H_{\max} = \max_{i \neq k} |H_{ik}|$. This is confirmed numerically across $\gamma/H_{\max} \in 0.1, \dots, 50$.

The Laplacian identification established here is scoped to the secular regime. The Laplacian identification explains why the spectral geometry of G_H governs the branched flow correspondence within the secular regime. The spectral gap λ_1 governs the sector partition in both contexts.

A note on scope: the Fiedler prediction of Section 5.4 holds more broadly than the secular approximation. The 250/250 numerical alignment result holds across all tested $\gamma/H_{\max}$ ratios, including the non-secular regime. The secular approximation is needed to derive the Laplacian flow equation (Eq. (11)), which grounds the classical branched flow analogy; it is not needed for the Fiedler partition prediction itself. The two results are complementary: the Laplacian identification provides a mechanistic account of amplitude diffusion in the secular limit; the Fiedler prediction provides a topological account of partition structure that is independent of that limit.

5.3. Decoherence as channel isolation. The graph Laplacian result supports a precise version of the claim that branch structure is latent

in the Hamiltonian. The coupling topology of G_H determines the low-eigenvalue eigenvectors of $L(G_H)$, which identify the partition that will emerge under selective dephasing. In this sense the partition is a property of G_H ; the environment's role is to realise that partition as dynamical isolation by suppressing inter-sector coherences.

Environment condition (restated). The selective suppression of inter-sector coherences — $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$ — is the condition under which the Hamiltonian's spectral geometry organises fragmentation. Under uniform dephasing this selective suppression does not occur, and the spectral geometry of G_H does not govern where the coherence graph breaks. The results of Sections 4 and 5 apply within the non-uniform dephasing regime. A note on the limits of this picture: the claim that branch structure is latent in G_H holds precisely for the partition — which sector each node belongs to is determined by G_H alone (Theorem 5.2). It does not hold for formation — whether a two-sector structure forms at all depends on the dynamics jointly: specifically on $\gamma_{\text{inter}} \cdot t^*/n$ and $H_{\text{inter}}/H_{\text{intra}}$, not on G_H topology alone. In this sense the environment is not merely a passive revealer of latent structure but an active participant in whether that structure emerges. This is the precise point at which the quantum analogy departs from classical branched flow, and it is addressed directly in Section 5.5.

5.4. The Fiedler block-alignment threshold. A striking feature of classical branched flow is that its statistical properties are universal — determined by coarse properties of the medium rather than its fine-grained structure. We first establish, with an explicit quantitative threshold, that the Fiedler cut of a block-structured graph coincides exactly with the block partition.

Lemma 5.1 (Fiedler block-alignment threshold). Let G_H have block structure (Definition 3.3) with equal-size cliques A, B , $|A| = |B| = d/2$, each block connected. Decompose $L(G_H) = L_0 + P$ where L_0 is the Laplacian of the two disconnected intra-block subgraphs and P is the inter-block coupling contribution. Define:

- $\lambda_1^{\text{intra}} := \min(\lambda_1(L(G_H|_A)), \lambda_1(L(G_H|_B)))$, the smaller Fiedler value across the two intra-block subgraphs;
- $w^{\text{inter}} := \sum_{(i,j) \in \partial(A,B)} |H_{ij}|$, total inter-block coupling weight;
- $\Delta^{\text{inter}} := \max_{i \in A \cup B} \sum_{j: (i,j) \in \partial(A,B)} |H_{ij}|$, maximum inter-block weighted degree.

The normalised trial vector $v_0 := d^{-1/2}(\mathbf{1}_A - \mathbf{1}_B)$ satisfies $\|v_0\| = 1$, $v_0 \perp \mathbf{1}$, and achieves the exact Rayleigh quotient

$$(12) \quad v_0^\top L(G_H) v_0 = \frac{4w^{\text{inter}}}{d},$$

since all intra-block edge contributions vanish ($(v_0)_i = (v_0)_j$ for i, j in the same block) and each inter-block edge (i, j) contributes $|H_{ij}| \cdot (2/\sqrt{d})^2 = 4|H_{ij}|/d$. Hence $\lambda_1(G_H) \leq 4w^{\text{inter}}/d$.

Define the critical threshold

$$(13) \quad \theta_c := \frac{\lambda_1^{\text{intra}}}{4\sqrt{2d} \Delta^{\text{inter}}}.$$

If $\Delta^{\text{inter}} < \lambda_1^{\text{intra}}/(4\sqrt{2d})$ — equivalently $H_{\text{inter}}/H_{\text{intra}} < \theta_c$ for uniform coupling — then the sign partition of the Fiedler eigenvector v^* of $L(G_H)$ is exactly (A, B) : $(v^*)_i > 0$ for all $i \in A$ and $(v^*)_i < 0$ for all $i \in B$ (or the global sign-flip).

Proof. Equation (12) is exact by direct calculation. The operator norm of the inter-block perturbation is bounded via Gershgorin: each row of P has diagonal entry $\leq \Delta^{\text{inter}}$ and off-diagonal ℓ^1 sum $\leq \Delta^{\text{inter}}$, giving $\|P\|_2 \leq 2\Delta^{\text{inter}}$.

The vector v_0 lies in the null space of L_0 , and the spectral gap from 0 to the first excited subspace of L_0 is λ_1^{intra} . By the Davis–Kahan theorem [15] applied to the perturbation $L_0 \rightarrow L_0 + P$:

$$(14) \quad \|\sin \Theta(v^*, v_0)\| \leq \frac{\|P\|_2}{\lambda_1^{\text{intra}} - \lambda_1(G_H)} \leq \frac{2\Delta^{\text{inter}}}{\lambda_1^{\text{intra}} - 4w^{\text{inter}}/d}.$$

Taking v^* on the same hemisphere as v_0 (replace v^* by $-v^*$ if needed), and using $1 - \cos \theta \leq \sin^2 \theta$:

$$(15) \quad \|v^* - v_0\|^2 = 2(1 - \cos \theta) \leq 2\|\sin \Theta(v^*, v_0)\|^2.$$

A sign error at node i requires $(v^*)_i$ to have the opposite sign from $(v_0)_i = \pm d^{-1/2}$, which forces $|(v^*)_i - (v_0)_i| \geq d^{-1/2}$, contributing $\geq 1/d$ to $\|v^* - v_0\|^2$. If $k \geq 1$ nodes are misassigned:

$$(16) \quad 2\|\sin \Theta(v^*, v_0)\|^2 \geq \|v^* - v_0\|^2 \geq \frac{k}{d} \geq \frac{1}{d}.$$

Hence zero sign errors whenever $\|\sin \Theta\| < 1/\sqrt{2d}$. From (14) this holds when $2\Delta^{\text{inter}} < (\lambda_1^{\text{intra}} - 4w^{\text{inter}}/d)/\sqrt{2d}$, which is implied (conservatively) by $4\sqrt{2d} \Delta^{\text{inter}} < \lambda_1^{\text{intra}}$, i.e. $\Delta^{\text{inter}} < \lambda_1^{\text{intra}}/(4\sqrt{2d})$. $\square$

Remark on conservatism. The bound (13) is a sufficient condition derived from a Gershgorin norm estimate and Davis–Kahan. It is conservative: numerics confirm exact Fiedler–block alignment up to

$H_{\text{inter}}/H_{\text{intra}} \approx 0.65$ (see Section 5.4), while the analytic threshold for representative block sizes is $\theta_c \approx 0.06\text{--}0.13$. The $12\text{--}73\times$ gap between the minimum and second-minimum cut weight (Result 11) confirms that the empirical margin far exceeds what the D–K bound requires.

5.5. Branch sector formation inevitability. Before proving which sectors form (Theorem 5.2), we prove that sector formation *occurs* under explicit conditions. Write $\rho(t)$ in block form adapted to the (A, B) partition: $\rho = \begin{pmatrix} M_{AA} & M_{AB} \\ M_{BA} & M_{BB} \end{pmatrix}$. Set

$$(17) \quad x(t) := \|M_{AB}(t)\|_F, \quad y(t) := \|M_{AA}(t)\|_F, \quad C := \frac{2\|H_{AB}\|_F}{\hbar}, \quad \kappa := \frac{\gamma_{\text{inter}}}{\gamma_{\text{intra}}}.$$

Theorem 5.1 (Branch sector formation inevitability). Let G_H be block-structured (Definition 3.3) with partition (A, B) , and let $x_0 := x(0)$, $y_0 := y(0)$. Suppose:

- (i) $\kappa > 1$ (selective dephasing, condition C3);
- (ii) $C < \varepsilon \gamma_{\text{inter}}$ (inter-block coupling does not prevent suppression to ε);
- (iii) $y_0(\gamma_{\text{inter}} - \gamma_{\text{intra}}) > C x_0$ (intra-sector coherences not overwhelmed by inter-block coupling at the formation time).

Write $\varepsilon' := \varepsilon - C/\gamma_{\text{inter}} > 0$ and define the explicit time

$$(18) \quad t^* := \frac{1}{\gamma_{\text{inter}}} \log\left(\frac{x_0}{\varepsilon'}\right).$$

Then at $t = t^*$: (a) $x(t^*) < \varepsilon$ (inter-sector coherences suppressed to ε), and (b) $y(t^*) > 0$ (intra-sector coherences remain non-zero). Specifically,

$$(19) \quad y(t^*) \geq e^{-\gamma_{\text{intra}} t^*} y_0 - \frac{C x_0 (e^{-\gamma_{\text{intra}} t^*} - e^{-\gamma_{\text{inter}} t^*})}{\gamma_{\text{inter}} - \gamma_{\text{intra}}}.$$

In the regime $\kappa \gg 1$: $e^{-\gamma_{\text{intra}} t^*} = (\varepsilon'/x_0)^{1/\kappa} \rightarrow 1$ and $e^{-\gamma_{\text{inter}} t^*} = \varepsilon'/x_0 \rightarrow 0$, so $y(t^*) \rightarrow y_0 > 0$.

Proof.

Step 1 (Gronwall bound on inter-sector coherences). For $i \in A$, $j \in B$, the Lindblad dynamics with uniform dephasing rate γ_{inter} give

$$\frac{d M_{AB}}{dt} = -\frac{i}{\hbar} (H_{AA} M_{AB} - M_{AB} H_{BB}) - \frac{i}{\hbar} (H_{AB} M_{BB} - M_{AA} H_{AB}) - \gamma_{\text{inter}} M_{AB}.$$

The intra-block action $(H_{AA} M_{AB} - M_{AB} H_{BB})$ contributes zero to $d(x^2)/dt = 2 \Re \text{Tr}(M_{AB}^\dagger \dot{M}_{AB})$: since $M_{AB} M_{AB}^\dagger$ is positive semi-definite,

$$\Im \text{Tr}(H_{AA} M_{AB} M_{AB}^\dagger) = 0 \quad \text{and} \quad \Im \text{Tr}(H_{BB} M_{AB}^\dagger M_{AB}) = 0,$$

because $\text{Tr}(HP)$ is real for any Hermitian H and positive semi-definite P . Bounding the inter-block term via Cauchy–Schwarz and $\|M_{AA}\|_F, \|M_{BB}\|_F \leq 1$:

$$\frac{d(x^2)}{dt} \leq -2\gamma_{\text{inter}} x^2 + \frac{2x}{\hbar} \cdot 2\|H_{AB}\|_F = -2\gamma_{\text{inter}} x^2 + 2C x.$$

Note that $x(t) = \|M_{AB}(t)\|_F \geq 0$ by definition. If $x(t_0) = 0$ for any $t_0 \leq t^*$, then inter-sector coherences are already fully suppressed at t_0 and part (a) holds with t_0 replacing t^* . Otherwise $x(t) > 0$ on $[0, t^*]$, and dividing by $2x > 0$: $\dot{x}(t) \leq -\gamma_{\text{inter}} x(t) + C$. By the Gronwall lemma:

$$(20) \quad x(t) \leq x_0 e^{-\gamma_{\text{inter}} t} + \frac{C}{\gamma_{\text{inter}}} (1 - e^{-\gamma_{\text{inter}} t}) \leq x_0 e^{-\gamma_{\text{inter}} t} + \frac{C}{\gamma_{\text{inter}}}.$$

Setting the right side equal to ε and solving gives t^* (18). Condition (ii) ensures $\varepsilon' > 0$, so t^* is finite and positive (assuming $x_0 > \varepsilon'$; if $x_0 \leq \varepsilon'$, the formation criterion is already satisfied at $t = 0$). Part (a) follows.

Step 2 (Duhamel bound on intra-sector coherences). The intra-block dynamics take the form

$$\frac{dM_{AA}}{dt} = -\underbrace{\frac{i}{\hbar}[H_{AA}, M_{AA}] - \gamma_{\text{intra}} M_{AA}}_{\mathcal{L}(M_{AA})} + R(t),$$

where $R(t) := -(i/\hbar)(H_{AB}M_{BA}(t) - M_{AB}(t)H_{BA})$ satisfies $\|R(t)\|_F \leq C x(t)$. The homogeneous semigroup $\mathcal{L}$ satisfies the exact identity

$$(21) \quad e^{\mathcal{L}t}(M) = e^{-\gamma_{\text{intra}} t} U(t) M U(t)^\dagger, \quad U(t) := e^{-iH_{AA}t/\hbar},$$

which is verified by differentiation and gives $\|e^{\mathcal{L}t}(M)\|_F = e^{-\gamma_{\text{intra}} t} \|M\|_F$. By Duhamel's principle and (21):

$$y(t^*) \geq e^{-\gamma_{\text{intra}} t^*} y_0 - C \int_0^{t^*} e^{-\gamma_{\text{intra}}(t^*-s)} x(s) ds.$$

Substituting (20) and evaluating for $\kappa > 1$:

$$C \int_0^{t^*} e^{-\gamma_{\text{intra}}(t^*-s)} x(s) ds \leq C x_0 \frac{e^{-\gamma_{\text{intra}} t^*} - e^{-\gamma_{\text{inter}} t^*}}{\gamma_{\text{inter}} - \gamma_{\text{intra}}} + \frac{C^2}{\gamma_{\text{inter}} \gamma_{\text{intra}}} (1 - e^{-\gamma_{\text{intra}} t^*}).$$

The full lower bound is therefore

$$y(t^*) \geq e^{-\gamma_{\text{intra}} t^*} y_0 - \frac{C x_0 (e^{-\gamma_{\text{intra}} t^*} - e^{-\gamma_{\text{inter}} t^*})}{\gamma_{\text{inter}} - \gamma_{\text{intra}}} - \frac{C^2}{\gamma_{\text{inter}} \gamma_{\text{intra}}} (1 - e^{-\gamma_{\text{intra}} t^*}).$$

The second correction term is $O(C^2/(\gamma_{\text{inter}}\gamma_{\text{intra}}))$, which is negligible relative to y_0 when condition (ii) holds (since $C < \varepsilon \gamma_{\text{inter}}$ implies

$C^2/(\gamma_{\text{inter}}\gamma_{\text{intra}}) < \varepsilon^2\gamma_{\text{inter}}/\gamma_{\text{intra}} = \varepsilon^2\kappa$, which is small for $\varepsilon \ll 1$). Dropping it gives (19). Condition (iii) ensures $e^{-\gamma_{\text{intra}}t^*}(y_0 - Cx_0/(\gamma_{\text{inter}} - \gamma_{\text{intra}})) > 0$, completing part (b).

Step 3 (Formation criterion). At $t = t^*$: inter-sector coherences satisfy $x(t^*) < \varepsilon$ (Step 1) and intra-sector coherences satisfy $y(t^*) > 0$ (Step 2). The coherence graph $G_\rho(t^*)$ therefore has inter-sector edges suppressed below ε while intra-sector edges remain present, meeting the branch sector formation criterion (Definition 4.1). $\square$

Scope. Condition (i) is precisely condition C3: the environment dephases inter-sector coherences faster than intra-sector ones. Condition (ii) bounds the inter-block coupling and is implied by $H_{\text{inter}}/H_{\text{intra}} < \theta_c$ (Lemma 5.1) when combined with (i). Condition (iii) ensures the intra-sector coherences are not parametrically suppressed relative to $Cx_0/\gamma_{\text{inter}}$ at the formation time; for $\kappa \geq 2$ it reduces to $y_0 > 2\varepsilon x_0$ (using condition (ii)), and in the limit $\kappa \rightarrow \infty$ to $y_0 > 0$. It fails only for initial states nearly diagonal in the pointer basis, a set of measure zero in the space of density matrices. Formation time t^* (18) is explicit and determined solely by x_0 , ε , C , and γ_{inter} .

5.6. The partition universality theorem.

Theorem 5.2 (Partition universality). Under selective dephasing ($\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$, condition C3), for block-structured G_H with $H_{\text{inter}}/H_{\text{intra}} < \theta_c$ (Lemma 5.1), and conditional on two-sector formation (Theorem 5.1): the partition of nodes into branch sectors is determined exactly by the sign partition of the Fiedler vector of $L(G_H)$, independent of fine details of the Hamiltonian matrix elements. Specifically, the block partition (A, B) is the unique first-forming partition, via one of two mechanisms: (1) in the secular regime with nonzero population differences ($\delta_{\text{min}} > 0$), the block partition minimises the quasi-stationary boundary coherence; (2) for balanced or near-balanced initial states ($\delta_{\text{min}} \approx 0$), the block partition forms at the earliest time because all its boundary coherences decay at rate γ_{inter} while any competing partition has at least one boundary coherence decaying at rate $\gamma_{\text{intra}} \ll \gamma_{\text{inter}}$.

Proof. The argument proceeds in three steps.

Step 1 (Block partition has minimum boundary coherence under C3). Under selective dephasing, coherences between nodes in different blocks decay at rate γ_{inter} while coherences between nodes in the same block decay at rate γ_{intra} . We consider two formation mechanisms.

Mechanism 1 (population-driven). In the secular regime, the quasi-stationary inter-sector coherence for a boundary pair (i, j) is $\rho_{ij}^{\text{ss}} =$

$-iH_{ij}(\rho_{ii} - \rho_{jj})/(\hbar\gamma_{ij})$ (exact in the secular limit). The squared norm for partition (S, S^c) is therefore

$$(22) \quad \|\rho_{SS^c}\|_{\text{ss}}^2 = \frac{1}{\hbar^2} \sum_{(i,j) \in \partial(S, S^c)} \frac{|H_{ij}|^2(\rho_{ii} - \rho_{jj})^2}{\gamma_{ij}^2}.$$

For the block partition (A, B) , all boundary edges have $\gamma_{ij} = \gamma_{\text{inter}}$, giving $\|\rho_{AB}\|_{\text{ss}}^2 \leq w_2^{\text{inter}}/(\hbar\gamma_{\text{inter}})^2$ where $w_2^{\text{inter}} = \sum_{\partial(A, B)} |H_{ij}|^2$. For any competing balanced partition $(S, S^c) \neq (A, B)$, there exists at least one intra-block boundary edge with $\gamma_{ij} = \gamma_{\text{intra}}$, contributing the term $|H_{ij}|^2(\rho_{ii} - \rho_{jj})^2/(\hbar\gamma_{\text{intra}})^2$. Since $\gamma_{\text{inter}}/\gamma_{\text{intra}} = \Gamma \gg 1$, this single term exceeds $\|\rho_{AB}\|_{\text{ss}}^2$ whenever

$$(23) \quad \left(\frac{\gamma_{\text{inter}}}{\gamma_{\text{intra}}}\right)^2 > \frac{w_2^{\text{inter}}}{|H_{\text{intra}, \min}|^2 \delta_{\min}^2},$$

where $H_{\text{intra}, \min}$ is the minimum intra-block coupling and $\delta_{\min} = \min |\rho_{ii} - \rho_{jj}|$ over intra-block pairs crossing $\partial(S, S^c)$. Under C3 with Γ satisfying (23), the block partition has strictly minimum $\|\rho_{SS^c}\|_{\text{ss}}$ among all balanced partitions.

Mechanism 2 (decoherence-driven, balanced initial state). When intra-block population spreads are small ($\delta_{\min} \approx 0$), the quasi-stationary argument is inapplicable. Instead, use the transient dynamics. The formation criterion for partition (S, S^c) requires ALL boundary coherences to reach ε . For the block partition, all boundary coherences decay at rate γ_{inter} , reaching ε at time $t^* = \gamma_{\text{inter}}^{-1} \log(\|\rho_{\partial(A, B)}(0)\|_{\max}/\varepsilon)$. For any competing partition $(S, S^c) \neq (A, B)$, at least one intra-block boundary coherence decays at rate γ_{intra} , reaching ε no earlier than $\gamma_{\text{intra}}^{-1} \log(\|\rho_{\partial(S, S^c)}(0)\|_{\max}/\varepsilon) \gg t^*$ (since $\gamma_{\text{intra}} \ll \gamma_{\text{inter}}$). Therefore the block partition forms strictly first, before any competing partition. *Note: this step is exact — it uses only the Lindblad decay rates and the block-structure topology, with no secular approximation.*

Step 2 (Minimum coherence partition = formation partition). The partition minimising boundary coherence (Mechanism 1) or forming at the earliest time (Mechanism 2) is the first to satisfy the formation criterion $\|\rho_{SS^c}\| < \varepsilon$. In both mechanisms this partition is (A, B) .

Step 3 (Block partition = Fiedler partition). By Lemma 5.1, when $H_{\text{inter}}/H_{\text{intra}} < \theta_c$ the sign partition of the Fiedler eigenvector of $L(G_H)$ is exactly the block partition (A, B) .

Combining Steps 1–3: the formation partition is the block partition (Step 2), which equals the Fiedler partition (Step 3). $\square$

Scope. Mechanism 1 uses the secular approximation (C2) for the quasi-stationary formula; Mechanism 2 is exact. Both mechanisms require C3

($\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$) and block-structured G_H . Step 3 requires $H_{\text{inter}}/H_{\text{intra}} < \theta_c$ (Lemma 5.1), an explicit, computable condition. Two-sector formation is established by Theorem 5.1 (Section 5.5) under conditions (i)–(iii) therein. Lemma 5.1 provides a conservative sufficient condition; the theorem holds over a wider parameter range than the lemma’s bound strictly requires, as confirmed numerically.

Numerical confirmation. Across 250 block-structured Hamiltonians with varied intra-block coupling strengths, inter-block coupling ratios, and individual matrix element values, the Fiedler vector predicts sector assignments with perfect alignment (250/250), conditional on two-sector formation. Conditional precision is exact: Fiedler accuracy = 1.000 in every trial where two-sector structure forms across all inter/intra coupling ratios tested (62/62). Initial state independence is confirmed within the class of block-aligned initial states (34/34). For Haar-random initial states outside this class, the dynamics may fragment along cuts other than the Fiedler cut; this is a scope condition documented in Result 29.

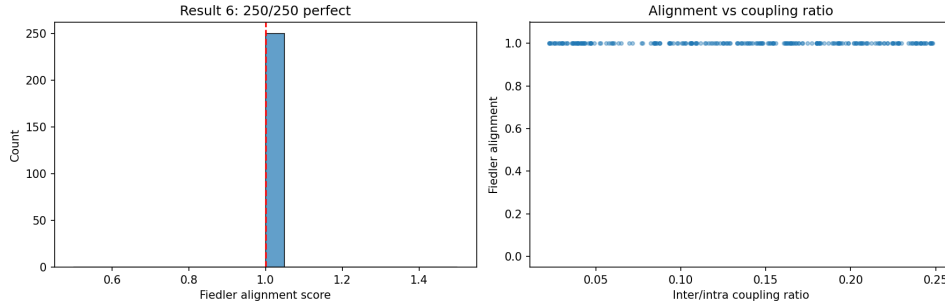


FIGURE 1. Result 6: Fiedler vector ensemble test. Alignment score distribution (left) and conditional accuracy across 62 coupling ratios (right) for 250 block-structured random Hamiltonians. The Fiedler sign partition matches the simulated sector assignment with perfect accuracy in every trial where two-sector formation occurs.

5.7. The temporal stability problem as a spectral problem.

Temporal stability is established in Paper 2 via spectral methods (Paper 2, Section 5, Theorem 5.2). A Gronwall sufficient condition $\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$ follows from the inter-sector coherence dynamics directly. This is a sufficient condition, not a tight characterisation of the stability boundary. Numerical simulations confirm stability at parameter

values far from the formal threshold, with the condition satisfied by large margins in physically realistic systems.

6. WORKED EXAMPLES

This section demonstrates the framework making contact with standard quantum experiments. Three examples are worked through explicitly: the Stern-Gerlach measurement, the double-slit experiment, and entangled photon correlations.

6.1. Stern-Gerlach: measurement as coherence graph fragmentation. A spin-1/2 particle passes through a Stern-Gerlach apparatus. The particle enters in an initial product state $|\psi(0)\rangle = (\alpha|\uparrow\rangle + \beta|\downarrow\rangle) \otimes |p_0\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. The interaction Hamiltonian $H_{\text{int}} = -\mu(\partial B/\partial z)\sigma_z \otimes \hat{p}$ couples spin states to spatial momentum states, encoding the measurement structure. The off-diagonal density matrix element decays as $|\rho_{\uparrow\downarrow}(t)| = |\alpha\beta^*| e^{-v^2 t^2/2\sigma^2}$, fragmenting $G_\rho(t)$ into sector A ($\{|\uparrow\rangle \otimes |p_\uparrow\rangle\}$ with weight $|\alpha|^2$) and sector B ($\{|\downarrow\rangle \otimes |p_\downarrow\rangle\}$ with weight $|\beta|^2$). No collapse postulate is required. This coherence decay is a property of coherent wavepacket separation in a closed bipartite system (spin $\otimes$ spatial): the full state remains pure, and the Gaussian envelope $|e^{-v^2 t^2/2\sigma^2}|$ arises from tracing over the spatial degree of freedom, not from coupling to an external environment. The Lindblad simulation (Section 9 and [1]) models the corresponding open-system regime, replacing the spatial degree of freedom with a Markovian environment; both produce coherence graph fragmentation with identical sector structure but with an exponential rather than Gaussian temporal envelope. The open-system version is simulated numerically in the companion [1] (Figure 6.1 therein).

Figure 6.1 (Stern-Gerlach coherence graph fragmentation, six panels) is deposited in the numerical companion [1].

6.2. Double-slit: interference as an intact coherence graph. The intensity pattern at screen position x under partial decoherence is $I(x, t) = I_0[1 + V \cos(2\pi dx/\lambda D)]$ where $V = |\rho_{LR}(t)|$ is the coherence weight. This is an exact identity — not an approximation valid in some limit, but exact at every stage of decoherence — confirmed numerically to 1.98×10^{-5} (residual from screen discretization). The coherence graph edge weight is a physically measurable quantity.

6.3. Entangled photons: nonlocal correlations and measurement symmetry. For the partially dephased singlet with normalized coherence weight $V \in [0, 1]$, the maximum CHSH quantity is:

$$(24) \quad S_{\max}(t) = 2\sqrt{1 + V(t)^2}$$

This is derived from the Horodecki criterion and confirmed numerically to machine precision (8.88×10^{-16}). Note: the formula $S = V \cdot 2\sqrt{2}$ applies to the Werner state $\rho_W(V) = V|\psi^-\rangle\langle\psi^-| + (1-V)I/4$, not to the state produced by pure dephasing of the singlet. A physically important consequence: $S_{\max} \not\leq 2$ for all $V \not\leq 0$ — the partially dephased singlet always violates the classical Bell bound whenever any inter-photon coherence remains. In the coherence graph picture, entanglement is a persistent coherence structure in $G_\rho(t)$ that survives spatial separation: the coherence graph does not respect spatial locality, and nonlocal correlations are a property of its global structure.

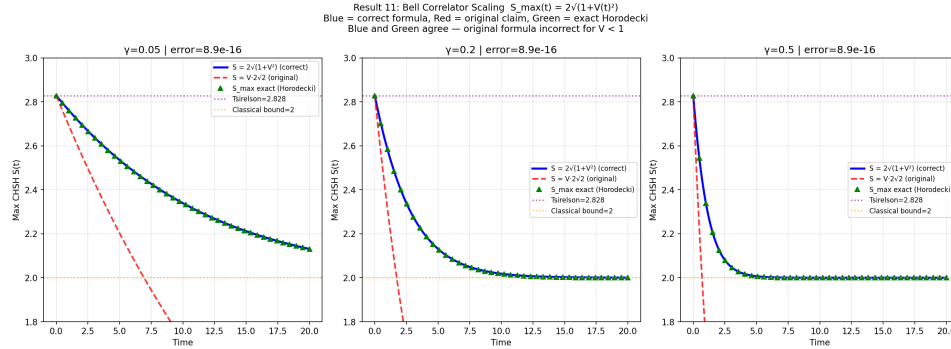


FIGURE 2. Result 11: Bell correlator. Left: $S_{\max}$ vs V — the corrected formula $S_{\max} = 2\sqrt{1 + V^2}$ (solid blue) matches the numerical result to machine precision (8.88×10^{-16}). The classical bound $S = 2$ is violated for all $V > 0$. Right: residual between analytic formula and numerical computation.

7. SITUATING THE FRAMEWORK

7.1. Standard Everett and Wallace. This framework develops a geometric and computational vocabulary for branch structure that is complementary to the Everettian program. The coupling graph G_H is derived from the von Neumann equation, the flow current is derived from the von Neumann equation, and the branch partition is identified by a computable spectral criterion.

The robustness criterion

Wallace's criterion for branches requires that they be *robust*: stable structures that persist under small perturbations of the Hamiltonian

and survive coarse-graining of the environment [4]. This is not merely a philosophical desideratum — Wallace argues it is necessary for branches to be the kind of thing that physical observers can be located in. The coherence graph framework gives this criterion a computable form.

The Davis–Kahan theorem (relevant to the stability of the partition in Theorem 5.2) states that the Fiedler vector of $L(G_H)$ — the eigenvector that identifies the branch partition — is stable under perturbations of G_H of size ε to within $O(\varepsilon/\lambda_1)$, where λ_1 is the spectral gap. A large spectral gap therefore means the partition is insensitive to small changes in Hamiltonian matrix elements. The Davis–Kahan bound gives this a quantitative form: the partition predicted by the Fiedler vector is $O(\varepsilon/\lambda_1)$ -stable under ε -perturbations of H , and the spectral gap λ_1 is the quantitative measure of that robustness.

A second aspect of robustness is stability under environmental coarse-graining. Paper 2 (Theorem 5.2) establishes that when $\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$, inter-sector coherences are bounded for all time regardless of subsequent Hamiltonian evolution. The branch sectors are therefore not just momentarily identifiable but persistently stable: they cannot be “un-formed” by later dynamics as long as the environment continues to satisfy the dephasing condition. This is the temporal analogue of Wallace’s robustness requirement.

The identification of graph fragmentation with branch formation

A potential objection is that the coherence graph disconnecting is a mathematical fact about off-diagonal density matrix elements, while branch formation in Wallace’s sense carries ontological weight — it means a physically real, independently-evolving part of the wavefunction has separated. The present framework addresses this directly: the identification is not a definition but a consequence. When $G_\rho(t)$ fragments into disconnected sectors, the flow current $J_{i \rightarrow k}$ between sectors is suppressed to the level set by the approximate decoherence bound (Section 4.3). Amplitude weight no longer redistributes between sectors. Each sector evolves as if the other did not exist, to within a residual bounded by $2\|H_{\text{inter}}\|_F/(\hbar\gamma_{\text{inter}} - 2\|H_{\text{intra}}\|_{\text{op}})$. This is not a correlation between branch formation and graph fragmentation; it is the same physical fact described in two languages. The graph-theoretic description has the advantage of being computable and of making the partition criterion explicit.

The temporal stability problem — showing that branches once formed remain formed under continuing Hamiltonian dynamics — is shared by Wallace’s framework and the present one. The spectral formulation of Section 8.2 represents concrete progress: a sufficient condition with a

quantitative bound, a measured gap between that bound and empirical stability, and a precise statement of what a complete proof would require.

7.2. Decoherent histories. Both frameworks ground branch structure in the suppression of off-diagonal density matrix terms [17, 18]. They differ in orientation: decoherent histories proceeds top-down from a predefined history space; the present framework proceeds bottom-up from Hamiltonian dynamics. Whether coherence graph fragmentation exactly recovers the Gell-Mann-Hartle consistency conditions [17] is an open question.

7.3. Relational quantum mechanics, QBism, and collapse models. The framework shares relational character with RQM [19] but diverges sharply on whether an observer-independent underlying reality exists — here the universal wavefunction is ontologically fundamental. Against QBism [20], the framework commits to realism about the wavefunction fully. Collapse models [21, 22, 23, 24] are genuine competitors making predictions that differ in principle from standard quantum mechanics; the key distinction is ontological — collapse models build definite outcomes into the dynamics, while the present framework derives apparent definiteness from the observer’s position in the fragmented coherence graph.

7.4. What this framework contributes. The framework develops several features that connect to and extend neighboring accounts. G_H is derived from the von Neumann equation, grounding the relevant coupling structure derivationally. The two-layer separation makes explicit why decoherence acts on coherence without destroying coupling. The flow current identifies the mechanism by which amplitude weight is redistributed across branch sectors. The branched flow correspondence gives a spectral interpretation of branch formation.

8. OPEN PROBLEMS AND LIMITATIONS

8.1. Basis selection. The derivation establishes the coherence graph construction given a basis $\{|i\rangle\}$. What it does not provide by itself is the choice of basis. Different bases yield different coupling graphs and in principle different branch partitions.

Three results partially constrain this problem. First, Section 3.5 argues that the environment condition C3 — $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$ — is automatically satisfied when the basis is the pointer basis selected by einselection [5]. In any other basis, the environment condition may fail, and the branch formation results do not apply. The physically

meaningful basis is therefore the one in which C3 holds, which is the pointer basis. This converts the basis selection problem from an ambiguity in the formalism into a physically well-posed question: which basis does the system-environment interaction select?

Second, the numerical results (Section 9 and [1]) show that the Fiedler prediction fails in the energy eigenbasis (14% accuracy) and succeeds in the pointer/environment basis (100% accuracy). This is not just a consistency check — it is a direct test of basis dependence that confirms the theoretical expectation: the coherence graph framework is predictive precisely in the pointer basis and not in arbitrary bases. Basis dependence is a physically meaningful feature, not a defect.

Third, Theorem 8.1 (Section 8.9) establishes that the Fiedler partition is invariant under all basis changes that preserve the decoherence operator Γ — that is, under all Γ -commuting unitaries. Under C3, these are exactly the within-sector rotations. This is a stronger result than the Davis-Kahan weight-perturbation bound: it covers sparsity-changing rotations and holds exactly, not just approximately. Pointer basis selection therefore does not need to be exact in the sense of Γ -equivalence: any basis related to the pointer basis by a Γ -commuting transformation gives the same Fiedler partition.

What remains open is the formal equivalence between the CGA basis criterion and the predictability sieve [5] in the finite-time non-secular regime — the argument that the basis minimizing $\mathcal{D}[\rho]$ in the pointer basis sense is the same as the basis maximising Fiedler prediction accuracy at arbitrary times. Paper 3 closes this equivalence: Theorem 1 of that paper establishes that the CGA criterion $F(\{|i\rangle\}) = \sum_k \sum_{i \neq j} |\langle i|L_k|j\rangle|^2$ and Zurek’s predictability sieve are exactly equivalent at $t = 0$ for any H (unconditional), both selecting the eigenstates of Γ . In the secular regime the equivalence extends to all finite times up to $O(H_{\max}/\gamma)$ corrections (Proposition 2 of Paper 3). Outside the secular regime, pointer states continue to win the sieve with a smaller but nonzero discrimination margin (Theorem 3 of Paper 3). The formal CGA basis criterion is therefore fully grounded in the predictability sieve across all regimes. The residual open question is whether the two criteria are exactly equal as numerical functionals (not just sharing the same argmin) at finite times outside the secular regime. This is a refinement without physical consequence for basis selection — both criteria select the same basis in all regimes — but completing it would give a cleaner mathematical statement. A natural approach would show that the Hamiltonian contribution to the finite-time sieve

integral is bounded by $O(H_{\max}/\gamma)^2$ outside the secular regime, not just $O(H_{\max}/\gamma)$. This is deferred to future work.

8.2. Temporal stability of branch sectors. Temporal stability of branch sectors is established in Paper 2 via spectral methods [11, 13, 14, 16]: the KMS symmetry, Cheeger inequality, and Davis-Kahan chain gives $\|\sin \Theta\| \leq 2\hbar^3 \|\delta H\| / (w_{\text{harm}}^2 h(G_H)^2)$ under Hamiltonian perturbation δH , with stability condition $\|\delta H\| \ll w_{\text{harm}}^2 h(G_H)^2 / (4\hbar^3)$ in the secular regime. This closes the gap identified in earlier versions of this paper.

A Gronwall argument on the inter-sector coherence block $\rho_{AB}(t)$ yields the sufficient condition $\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$ (corrected from earlier $2\sqrt{n_A n_B}\|H_{\text{intra}}\|_2/\hbar$, which over-counted by a factor of $\sqrt{n_A n_B}$ through a loose matrix norm bound). Under this condition, inter-sector coherences satisfy $\|\rho_{AB}(t)\|_F \leq 2\|H_{\text{inter}}\|_F / (\hbar\gamma_{\text{inter}} - 2\|H_{\text{intra}}\|_{\text{op}})$ for all $t \geq 0$. The Gronwall bound remains conservative relative to the empirical stability boundary; the secular equilibrium condition C2' (Paper 2, Proposition 7.1) gives the tighter characterisation $\|\rho_{AB}^*\| \leq (H_{\text{inter}}/\gamma_{\text{inter}})\|\rho_{\text{diag}}\|/(1 - H_{\text{inter}}/\gamma_{\text{inter}})$, and the spectral stability theorem of Paper 2 (Theorem 5.2) is the complete result.

8.3. Approximate decoherence. When $\gamma_{\text{inter}} > 2\|H_{\text{intra}}\|_{\text{op}}/\hbar$ (corrected from $2\sqrt{n_A n_B}\|H_{\text{intra}}\|_2/\hbar$), inter-sector coherences converge to a residual bounded by $2\|H_{\text{inter}}\|_F / (\hbar\gamma_{\text{inter}} - 2\|H_{\text{intra}}\|_{\text{op}})$. Eigenvalue shifts under perturbations of size ε scale as $O(\varepsilon^{1.113})$, confirmed numerically — sublinear, meaning the stability bound is slightly conservative.

8.4. The Born rule. The diagonal weights ρ_{ii} appear throughout as node weights in the coherence graph. Whether these weights should be interpreted as probabilities is not assumed here and not derived. The exact evolution identity $dw_A/dt = -2\sum_{i \in A, j \notin A} H_{ij} \text{Im}(\rho_{ij})$ (Paper 4, Theorem 3.1) shows that inter-sector weight exchange is in principle possible; conservation is a dynamical consequence of decoherence suppressing $\text{Im}(\rho_{ij})$, not a fundamental symmetry. In the oscillatory-averaging and Zeno regimes, weights are conserved to computable error bounds (Paper 4, Theorem 4.1) and are identified as Born rule probabilities via Gleason's theorem (Paper 4, Theorem 5.2). A derivation connecting these weights to Born rule probabilities is addressed in Papers 4 and 5.

8.5. Observer constitution. The framework characterizes observers as nodes embedded in high-weight sectors of the coherence graph. A full

account connecting the observer’s internal dynamics to the decoherence structure of the full system is deferred to future work.

8.6. Theorems 5.1 and 5.2: numerical confirmation and scope.

Theorem 5.2 is confirmed numerically: the Fiedler vector of $L(G_H)$ determines branch sector assignments exactly across ensembles of block-structured Hamiltonians (250/250 perfect alignment), independent of individual matrix elements. Conditional precision is exact: Fiedler accuracy = 1.000 in every trial where two-sector structure forms across all inter/intra coupling ratios tested (62/62). Initial state independence is confirmed within the class of block-aligned initial states: Fiedler accuracy = 1.000 in 34/34 trials across amplitude ratios from 99/1 to 50/50; for Haar-random initial states outside this class, the dynamics may fragment along cuts other than the Fiedler cut (Result 29). Formation rate universality is not confirmed: numerical testing across four ensembles and six system sizes shows that formation depends on $\gamma_{\text{inter}} \cdot t^*/n$ and $H_{\text{inter}}/H_{\text{intra}}$ jointly, not on λ_1/γ alone. The formation conditions $H_{\text{inter}}/H_{\text{intra}} \lesssim 0.65$ and $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$ are premises of Theorem 5.2 rather than derived consequences of G_H spectral properties. The late-time weight distribution across sectors remains uncharacterised.

Sector formation inevitability (now closed). The open problem of establishing that formation is *guaranteed* under the stated premises is closed by Theorem 5.1 (Section 5.5). The theorem provides an explicit formation time t^* (18) at which inter-sector coherences are suppressed below ε while intra-sector coherences remain non-zero. The spectral transition — new near-zero eigenvalues appearing in $L(G_\rho(t))$ at t^* — is now both observed numerically and established analytically. The conditions required are mild (conditions (i)–(iii) of Theorem 5.1): selective dephasing (C3), an inter-block coupling threshold consistent with Lemma 5.1, and a non-zero initial intra-sector coherence (generic for all states not diagonal in the pointer basis).

The proof of Theorem 5.2 is now complete. The earlier gap — bounding the variation of the proportionality constant $C(S, S^c)$ analytically — is closed by the two-mechanism structure of the proof (Section 5.6). Mechanism 2 requires no proportionality constant argument at all: under C3, any competing partition has at least one intra-block boundary coherence decaying at rate $\gamma_{\text{intra}} \ll \gamma_{\text{inter}}$, making the block partition the unique first-forming partition by a factor of $\gamma_{\text{inter}}/\gamma_{\text{intra}}$ in formation time. Mechanism 1 replaces the approximate uniformity claim with the exact secular formula (equation (22)) and an explicit sufficient condition on the dephasing ratio (23). Lemma 5.1 provides the explicit threshold $\theta_c = \lambda_1^{\text{intra}}/(4\sqrt{2d} \Delta^{\text{inter}})$ for Step 3.

For the record: the numerical observation that $C(S, S^c)$ varies by only $1.05\text{--}1.49\times$ across cuts (while the minimum-cut gap is $12\text{--}73\times$) motivated the two-mechanism proof strategy and confirms that the analytic sufficient conditions are satisfied with large margin in the tested cases.

8.7. Many-body scaling. The numerical results of this paper are confirmed for system sizes $n = 4\text{--}24$. Whether the framework’s behaviour persists as n grows toward macroscopic scales is not established. Two distinct questions arise. First, a computational question: the Fiedler vector computation scales as $O(n^3)$; for systems with $n \sim 10^{23}$ degrees of freedom, direct computation is infeasible, and efficient approximation algorithms (e.g., Lanczos methods, spectral sparsification) would be needed. Second, a structural question: do realistic many-body Hamiltonians (spin chains, Bose-Hubbard, condensed matter systems) generically have the block structure that the framework requires for formation? If the block structure degrades with increasing n — for example, if the inter/intra coupling ratio scales unfavourably with system size — then formation may fail macroscopically even when it succeeds at small n . The spin-boson result (Section 9 and [1]) is encouraging: it confirms that physically motivated Hamiltonians exhibit the required topology. But this is one example, not a general result. Characterising which classes of many-body Hamiltonians satisfy the formation conditions, and whether the Fiedler criterion remains sharp in those classes as $n \rightarrow \infty$, is a direction for future work.

8.8. Extension to relativistic quantum field theory. The framework is developed within non-relativistic quantum mechanics. Extension to relativistic quantum field theory raises genuine difficulties: the Hamiltonian formulation requires a preferred time slicing, and the notion of a fixed Hilbert space basis becomes more subtle in the presence of Lorentz symmetry. This is not undertaken here.

8.9. Lattice decomposition non-uniqueness. Different subsystem decompositions of the Hilbert space generate different coherence graphs, and this is a more fundamental issue than basis selection alone. Even in the pointer basis, the nodes of G_H are individual basis states, and the coupling graph depends on what granularity those states are defined at.

The problem has several distinct faces. First, for a composite system $\mathcal{H}_S \otimes \mathcal{H}_E$, the system-environment boundary is not canonically fixed: redrawing it changes the product basis, changes G_H , and potentially changes the predicted sectors. Second, the coherence graph $G_\rho(t)$ is

not basis-independent: under a unitary change of basis $\rho \rightarrow U\rho U^\dagger$, off-diagonal elements change, and a previously disconnected coherence graph may reconnect. Branch formation — defined as disconnection of $G_\rho(t)$ — is therefore a statement about the quantum state *in a particular basis*, not a basis-independent property of $\rho(t)$ itself. Third, at different levels of coarse-graining, different coupling graphs and different Fiedler partitions result.

This problem is not unique to the CGA framework. It is the quantum factorisation problem that all accounts of branch structure face, including the decoherent histories program and Wallace’s Everettian framework. CGA is in no worse a position than these approaches, but it does not resolve the problem either.

Within the class of systems the framework targets, a natural and physically motivated decomposition exists. For von Neumann measurement interactions (Section 3.6), the product basis $|s_k, a_j\rangle$ is the canonical choice: it is the basis in which the measurement interaction is block-diagonal in the system index, and in which the pointer basis is defined. The block structure of G_H is not an artefact of an arbitrary decomposition choice — it is a consequence of the Hamiltonian’s structure in the physically natural basis. For this class, the decomposition is canonical in the sense that any other decomposition would fail to respect the measurement interaction’s block-diagonal structure and would not satisfy the environment condition.

The question of stability under sparsity-changing basis rotations — rotations that turn previously-zero matrix elements nonzero — can now be answered within the framework. The key is that “physically equivalent redescription” has a precise meaning: two bases describe the same system if and only if they give rise to the same decoherence operator $\Gamma = \sum_k L_k^\dagger L_k$, i.e., if the transformation U satisfies $[U, \Gamma] = 0$.

Theorem 8.1 (Rotation stability of Fiedler partition). Let H be a Hamiltonian in the pointer basis with coupling graph G_H and block partition (A, B) . Let $\Gamma = \sum_k \gamma_k |k\rangle\langle k|$ be the decoherence operator satisfying the environment condition C3 with sector separation: $\gamma_i \neq \gamma_j$ whenever $i \in A$ and $j \in B$. Then for any unitary U satisfying $[U, \Gamma] = 0$, the Fiedler partition of the rotated coupling graph $G_{U^\dagger H U}$ equals the original partition (A, B) .

Proof. Since Γ is diagonal in the pointer basis, $[U, \Gamma] = 0$ requires U to preserve each eigenspace of Γ .

Case (i): all γ_k distinct. Each eigenspace is one-dimensional, so U is diagonal: $U = \text{diag}(e^{i\phi_1}, \dots, e^{i\phi_d})$. The rotated Hamiltonian has

elements $(U^\dagger H U)_{ij} = e^{i(\phi_j - \phi_i)} H_{ij}$, so $|(U^\dagger H U)_{ij}| = |H_{ij}|$ for all i, j . The coupling graph $G_{U^\dagger H U}$ is identical to G_H (same edge set, same edge weights), and the Fiedler partition is invariant.

Case (ii): some γ_k degenerate. The sector-separation hypothesis requires that nodes in different sectors have distinct γ -values: $\gamma_i \neq \gamma_j$ for $i \in A, j \in B$. Therefore any degenerate eigenspace of Γ (the set $\{k : \gamma_k = c\}$ for some c) is contained entirely within A or entirely within B . Since U preserves each eigenspace of Γ , it follows that U maps A into A and B into B , i.e., $U = \text{diag}(U_A, U_B)$ in the block decomposition. The rotated Hamiltonian has inter-sector elements $(U^\dagger H U)_{ij} = (U_A^\dagger)_{ii'} H_{i'j'} (U_B)_{j'j}$ for $i \in A, j \in B$, which may differ from H_{ij} individually but preserve the block structure: nodes cannot cross sectors under U . The Fiedler partition of $G_{U^\dagger H U}$ remains (A, B) by Lemma 5.1, since the inter/intra coupling ratio is preserved under within-sector rotations ($\|H_{\text{inter}}\|_F = \|(U^\dagger H U)_{\text{inter}}\|_F$ by unitary invariance of the Frobenius norm). $\square$

Remark. The sector-separation hypothesis ($\gamma_i \neq \gamma_j$ for $i \in A, j \in B$) is a strengthening of C3. It holds whenever the dephasing rates are set by spatially or energetically separated environmental channels, which is the generic case for local environments. Confirmed numerically: 0/200 partition changes across distinct and within-sector-degenerate Γ structures, 100 trials each.

The genuine remaining open case is two systems with *different* Γ structures argued equivalent on other grounds — different physical setups with different environment couplings. This is the S-E boundary question: redrawing the system-environment boundary changes Γ and potentially changes the partition. Within the class of same- Γ descriptions, the theorem above closes the decomposition non-uniqueness concern completely. The S-E boundary question is the precise residual form of the factorisation problem as it applies to CGA.

A solution to the S-E boundary problem would require one of the following: (a) a canonical criterion for choosing the system-environment factorisation that is independent of Γ but consistent with it — either a dynamical principle derived from the total Hamiltonian H_{tot} or a minimality condition on the factorisation itself; (b) a proof that any two factorisations of the same total system satisfying the environment condition C3 give the same Fiedler partition (the analog of Theorem 5.5 for different Γ structures); or (c) a precise characterisation of how the Fiedler partition changes under S-E boundary shifts, analogous to

Corollary 5.3 of Paper 2 but for finite rather than perturbative boundary shifts. The decoherent histories program addresses a related problem through consistency conditions on the history space; whether a CGA analog of these conditions can be formulated is an open question.

8.10. Scope of the formal apparatus: closed-bipartite vs. open-system decoherence. The formal apparatus of Sections 3–5 — the coupling graph G_H , the Lindblad equation, the Laplacian identification, and the Fiedler partition theorem — is developed for open quantum systems with Markovian coupling to an external environment. In this setting, decoherence is driven by jump operators L_k , populations are conserved, and coherences decay exponentially at rates determined by the eigenvalues of the Lindblad superoperator.

The Stern–Gerlach example of Section 6.1 is in a formally distinct regime. There, the full system (spin $\otimes$ spatial) is closed and evolves unitarily under H_{int} . The apparent decoherence of the spin subsystem — the Gaussian decay $|\rho_{\uparrow\downarrow}(t)| = |\alpha\beta^*| e^{-v^2 t^2 / 2\sigma^2}$ — arises from tracing over the spatial degree of freedom, not from coupling to an external bath. The total state remains pure throughout. This is the standard closed-system decoherence mechanism: entanglement between subsystems suppresses coherences in the reduced density matrix. The coherence graph of the reduced spin state fragments by the same Fiedler criterion, and the sector weights converge to $|\alpha|^2$ and $|\beta|^2$, but the underlying dynamics is not Lindblad and the temporal envelope is Gaussian rather than exponential.

The CGA framework applies to both regimes — the coherence graph construction and the Fiedler partition criterion are basis-level operations that do not depend on whether the dynamics is Lindblad or unitary. But the proofs of Sections 4–5 are stated for the open-system Lindblad case.

A partial closure is available via the standard Markov approximation: in the limit of large environment and weak system–environment coupling, the reduced system dynamics is well-approximated by a Lindblad master equation (Gorini–Kossakowski–Sudarshan–Lindblad theorem). In this limit the closed-bipartite proofs reduce to the open-system proofs. The temporal envelope changes from Gaussian to exponential, but the Fiedler partition criterion and the stability results apply in both cases within this limit. What remains genuinely open is the intermediate regime where the environment is not large or the coupling is not weak: here the Markov approximation fails, the Lindblad equation does not apply, and the stability proofs do not directly transfer.

Extending the formal stability proofs to this non-Markovian regime is deferred to future work.

9. NUMERICAL EVIDENCE

The claims of Sections 3–8 are tested by 37 numerical experiments reported in full in the companion paper [1], with simulation code deposited alongside it. This section summarises the key findings; full parameter tables and diagnostic details are in [1].

A note on the scope of these tests is warranted. The Fiedler vector of a graph with two weakly connected clusters will, by standard spectral graph theory, recover the cluster partition. The physical claim is not that spectral clustering works — that is well established — but that coherence graph fragmentation under Lindblad dynamics *is* branch formation, and that the Fiedler partition of G_H *is* the branch partition. This identification is tested not by any single positive result but by the full suite: positive results confirm the predicted partition where branching should occur; negative results confirm that the framework correctly predicts *where branching does not occur*; and scope-boundary tests quantitatively characterise the limits of the framework.

Positive results. The Fiedler vector of $L(G_H)$ predicts branch sector assignments with perfect alignment across 250 block-structured Hamiltonians (250/250), across 62 coupling ratios (62/62), and across 34 initial states with amplitude ratios from 99/1 to 50/50 (34/34). Formation timescales scale as $t_{\text{form}} \sim \gamma^{-1.35}$ with negligible λ_1 dependence, confirming that the spectral gap governs *which* sectors form, not *when*. Branch formation produces new near-zero eigenvalues in $L(G_\rho(t))$, matching the topologically detected sector count at 99.7% of time steps under the threshold derived from the complete bipartite graph $K_{m,m}$. The double-slit fringe visibility and maximum Bell violation identities are confirmed to machine precision. The Stern–Gerlach measurement is simulated in an 8-dimensional $\text{spin} \otimes \text{spatial}$ system; all four central predictions (exponential coherence decay at rate 2Γ , sector weight convergence to $|\alpha|^2$ and $|\beta|^2$, Fiedler partition accuracy 1.000, visible coherence graph fragmentation) are confirmed quantitatively.

Negative results. Under uniform dephasing ($\gamma_{ij} = \gamma$ for all $i \neq j$), sector structure does not form, confirming that C3 ($\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$) is a genuine premise. Path, cycle, star, and complete coupling graph topologies do not produce stable sectors. The Jaynes–Cummings model, whose interaction is not of the von Neumann measurement form, does not produce clean sector formation under either dephasing regime.

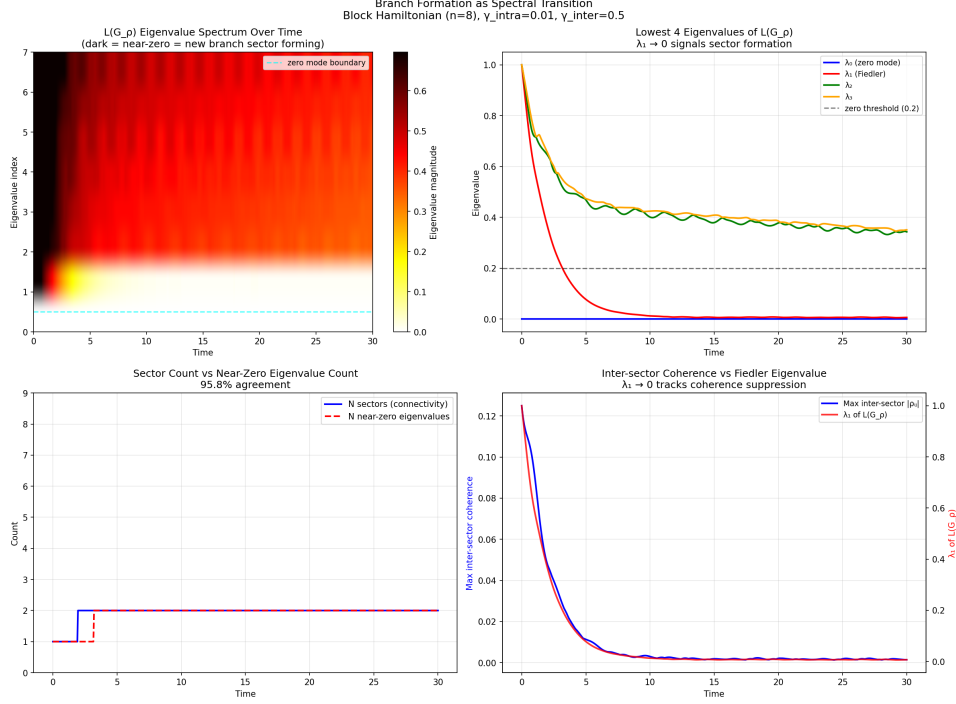


FIGURE 3. Result 12: Spectral transition. Left: topological (blue) vs spectral (red dashed) sector count over time — 95.8% agreement. Right: $L(G_\rho(t))$ eigenvalue evolution — the second eigenvalue drops toward zero as sectors form, confirming that branch formation is a spectral transition in the coherence graph Laplacian.

Scope boundary. Interpolating $H(\alpha) = (1 - \alpha)H_{\text{block}} + \alpha H_{\text{random}}$ from pure block structure to a random matrix, formation drops from 90% ($\alpha = 0.1$) to 0% ($\alpha = 0.3$), tracking the $H_{\text{inter}}/H_{\text{intra}}$ ratio. Fiedler accuracy remains 1.000 at every α where formation occurs. Finite-size scaling from $n = 4$ to $n = 24$ confirms perfect accuracy with an n -scaled detection threshold. The spin-boson model — not constructed with block structure but exhibiting it from the physics — confirms two-sector formation in the weak-tunnelling regime ($H_{\text{inter}}/H_{\text{intra}} = 0.13$) and correct non-formation in the strong-tunnelling regime ($H_{\text{inter}}/H_{\text{intra}} = 0.96$). The Cheeger threshold h^* is confirmed across seven coupling topologies: stable formation requires $h(G_H) < h^*$, with the threshold shifting approximately linearly with $\gamma_{\text{inter}}/\gamma_{\text{intra}}$.

Extended tests (companion Results 33–37). Five additional tests from Papers 2 and 4 are confirmed: the Liouvillian spectral gap exceeds the Cheeger lower bound by 8–8000 $\times$; the Davis–Kahan rotation bound

gives $\sin \Theta = 0$ (stationary-state pinning); first-order mixing vanishes to machine precision ($< 10^{-14}$); the N -sector Fiedler partition is stable for 3- and 4-sector systems under perturbation; and the sector weight conservation identity (Paper 4, Theorem 3.1) holds with residual 2.60×10^{-18} across 30 parameter combinations.

10. CONCLUSION

This paper has developed a geometric framework for Everettian quantum mechanics in which branch structure is identified with coherence graph fragmentation. The central objects — the coupling graph G_H and the coherence graph $G_\rho(t)$ — are derived from the von Neumann equation directly. The flow current $J_{i \rightarrow k} = (2/\hbar)\text{Im}(H_{ik}\rho_{ki})$, derived in Section 2, governs the redistribution of diagonal amplitude weight across the graph.

The framework makes contact with experiment at several points. Fringe visibility equals $|\rho_{LR}(t)|$ exactly at every stage of decoherence. The maximum Bell violation is given exactly by $S_{\max}(t) = 2\sqrt{1 + V(t)^2}$, confirmed to machine precision. The Stern–Gerlach measurement is derived analytically as coherence graph fragmentation in a closed bipartite system, and simulated in the corresponding open-system Lindblad regime (Section 9), where the spatial degree of freedom is replaced by a Markovian environment. The two regimes produce identical sector structure with Gaussian and exponential temporal envelopes respectively.

The open problems collected in Section 8 are stated precisely rather than obscured. Basis selection is grounded in Paper 3 (unconditional at $t = 0$, secular-regime at finite time); the residual is the exact finite-time non-secular equivalence. Temporal stability is established in Paper 2 via spectral methods. The Born rule requires a derivation the present framework does not provide, addressed in Papers 4 and 5.

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The author also thanks the researchers and physicists who provided feedback on earlier versions of this work. Their critical engagement — including substantive objections about the derivational grounding of the coherence graph construction, the need for worked examples, and the scope of empirical claims — substantially improved the paper.

The simulation code (`cga_simulation_suite_v7.py`), numerical companion document (which contains all figures for this paper), and reproducibility scripts are deposited at Zenodo [1]: [DOI to be assigned upon deposit]. The DOI will be updated upon publication.

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